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MIVA IFT 212

COMPUTER ARCHITECTURE AND ORGANIZATION



ASSESSMENT 1: SEQUENTIAL EXECUTION CYCLE CALCULATION

Topic: Non-Pipelined Performance Analysis

For the given instruction sequence, calculate the execution cycles for a non-pipelined (sequential) processor where each stage takes 1 cycle.

Instruction Sequence

1. LOADI R1, 10 R1 = 10
2. LOADI R2, 20 R2 = 20
3. ADD R3, R1, R2 R3 = R1 + R2
4. SUB R4, R3, R1 R4 = R3 - R1
5. STORE R4, 100 Mem[100] = R4
6. LOAD R5, 100 R5 = Mem[100]

Questions

Q1. Cycle Count Table (8 marks)

Complete the following table showing which instruction is in each stage for cycles 1-30:

Cycle	Fetch	Decode	Execute	Memory	Write-Back
1	I1	-	-	-	-
2	-	I1	-	-	-
3	-	-	I1	-	-
4	-	-	-	I1	-
5	-	-	-	-	I1
6					
…					
30					

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Q2. Total Cycles (3 marks)

How many total cycles does this program take? Show your calculation.

Q3. Final Register Values (4 marks)

Assuming all registers start at 0, what are the final values of:
- R1 = _____
- R2 = _____
- R3 = _____
- R4 = _____
- R5 = _____

Q4. Memory Value (2 marks)

What is the value stored at memory address 100? _____

Q5. CPI Calculation (3 marks)

What is the Cycles Per Instruction (CPI) for this program? Show your work.

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ASSESSMENT 2: PIPELINE DIAGRAM CONSTRUCTION

Topic: 5-Stage Pipeline Instruction Flow

Draw a complete pipeline diagram for the following 4-instruction program running on a 5-stage pipelined processor with no hazards.

Instruction Sequence

I1: LOADI R1, 5
I2: LOADI R2, 3
I3: ADD R3, R1, R2
I4: SUB R4, R3, R1

Questions

Q1. Pipeline Diagram (12 marks)

Complete the pipeline diagram for cycles 1-8:

Cycle	Fetch	Decode	Execute	Memory	Write-Back
1	I1	-	-	-	-
2	I2	I1	-	-	-
3					
4					
5					
6					

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7					
8					

Q2. Total Cycles (3 marks)

How many total cycles does this pipelined program take? _____

Q3. Formula Derivation (3 marks)

For a program with N instructions on a 5-stage pipeline, what is the formula for total cycles?

Formula: Cycles = _____

Q4. Stage Activity (2 marks)

In cycle 4, which instructions are in each stage?
- Fetch: _____
- Decode: _____
- Execute: _____
- Memory: _____
- Write-Back: _____

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ASSESSMENT 3: PIPELINE SPEEDUP CALCULATION

Topic: Performance Comparison

Calculate and analyze the speedup achieved by pipelining for programs of different lengths.

Given Data

Program	Instructions	Sequential Cycles	Pipelined Cycles
A	4	20	8
B	8	40	12
C	12	60	16
D	20	100	
E	100	500	

Questions

Q1. Complete the Table (4 marks)

Fill in the missing pipelined cycles for:
- Program D: _____ cycles
- Program E: _____ cycles

Q2. Speedup Calculation (8 marks)

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Calculate the speedup for each program:
- Speedup = Sequential Cycles / Pipelined Cycles

Program	Instructions	Speedup
A	4	
B	8	
C	12	
D	20	
E	100	

Q3. Analysis Questions (8 marks)

a) What trend do you observe in speedup as the number of instructions increases? (2 marks)

b) What is the theoretical maximum speedup for a 5-stage pipeline? Why? (2 marks)

c) For Program A (4 instructions), why is the speedup less than the maximum? (2 marks)

d) If Program E took 104 cycles instead of the ideal, what would its speedup be? (2 marks)

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ASSESSMENT 4: DATA HAZARD IDENTIFICATION

Topic: Recognizing RAW Hazards

Identify Read-After-Write (RAW) data hazards in given instruction sequences and determine how many stall cycles are needed.

Instruction Sequences

Sequence 1:
I1: ADD R1, R2, R3
I2: SUB R4, R1, R5
I3: ADD R6, R4, R7

Sequence 2:
I1: LOADI R1, 10
I2: LOADI R2, 20
I3: ADD R3, R1, R2
I4: STORE R3, 100
I5: LOAD R4, 100

Sequence 3:
I1: LOADI R1, 5
I2: ADD R2, R1, R3
I3: SUB R4, R2, R1
I4: ADD R5, R4, R6

Questions

Q1. Hazard Identification (9 marks)

For each sequence, list all RAW hazards in the format: "Ix reads register Ry before Iy writes it"

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Sequence	Hazards
1	
2	
3	

Q2. Stall Cycle Calculation (6 marks)

For a 5-stage pipeline with stalling, how many stall cycles are needed for each hazard?

Hazard	Stall cycles needed
I2 reads register from I1 in EX stage	
I2 reads register from I1 in MEM stage	
I2 reads register from I1 in WB stage	

Q3. Total Stall Cycles (3 marks)

For Sequence 3, how many total stall cycles will be inserted? Explain your reasoning.

Q4. Pipeline Stage Timing (2 marks)

In which pipeline stage are registers read? In which stage are they written?

- Registers read in: _____ stage
- Registers written in: _____ stage

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ASSESSMENT 5: HAZARD DETECTION LOGIC

Topic: Implementing Hazard Detection

Write pseudocode or Python code for hazard detection logic and determine when stalls are needed.

Given Pipeline State

At a particular cycle, the pipeline registers contain:

Register	Instruction	Destination	Valid
IF/ID	ADD R5, R1, R2	R5	True
ID/EX	SUB R4, R3, R1	R4	True
EX/MEM	LOADI R1, 10	R1	True
MEM/WB	ADD R2, R6, R7	R2	True

Questions

Q1. Hazard Detection Logic Pseudocode (8 marks)

Write pseudocode for a function “detect_raw_hazard(decoded_instr)” that:
1. Takes a decoded instruction (from IF/ID)
2. Checks its source registers against destinations in later stages
3. Returns True if a hazard exists, False otherwise

```
# Starter code
function detect_raw_hazard(decoded_instr):
    # Get source registers from decoded_instr
    src_regs = []
    if decoded_instr.src1 is not None:
        src_regs.append(decoded_instr.src1)
```

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```
    if decoded_instr.src2 is not None:
        src_regs.append(decoded_instr.src2)

    # Check against ID/EX (EX stage)
    if ID_EX.valid and ID_EX.dest_reg in src_regs:
        return True

    # Check against EX/MEM (MEM stage)
    _____

    # Check against MEM/WB (WB stage)
    _____

    return False
```

Q2. Hazard Detection for Given State (6 marks)

The instruction in IF/ID is “ADD R5, R1, R2”. Using the pipeline state above:

a) What are the source registers of this instruction? _____, _____

b) Check each later stage:

Stage	Register	Destination	Match?
EX	ID/EX	R4	No
MEM	EX/MEM	R1	-
WB	MEM/WB	R2	-

c) Is a hazard detected? _____

d) If yes, which register(s) cause the hazard? _____

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Q3. Stall Decision (3 marks)

If a hazard is detected:

- Should the pipeline stall? _____
- How many cycles will it stall? _____
- What is inserted into the pipeline during a stall? _____

Q4. Modified Detection (3 marks)

If we only checked against the EX stage (ID/EX), would we still detect this hazard? Why might this be insufficient?

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